

Branch Specialization in Attention

Phase 0 Raw Attention-Score Stability Checkpoint

Research checkpoint

May 21, 2026

Core framing:

Does structural branch design in transformer-style architectures induce stable functional specialization or functional modularity across random seeds?

Phase 0 target:

Before training new architectures, verify that cross-seed attention-head similarity can be measured cleanly on existing multi-seed models.

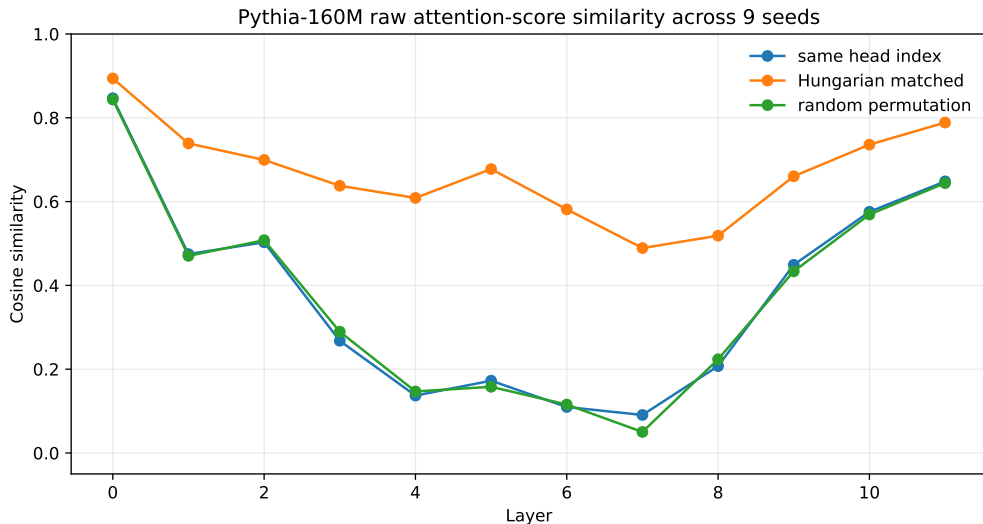
- Model: Pythia-160M, seeds 1–9, final revision step143000.
- Probe set: 8 short texts, max sequence length 64.
- Unit: layer-head attention matrix.
- Similarity: cosine similarity over valid causal token pairs.
- Alignment: raw same-index comparison and Hungarian-matched comparison.
- Null: random head permutation baseline within layer.

Key Aggregate Result

Metric	Probabilities	Raw scores
Same-index mean	0.7127	0.3735
Hungarian-matched mean	0.8127	0.6692
Matched minus random mean	0.0998	0.2982
Max layer-pair gap	0.2272	0.7550

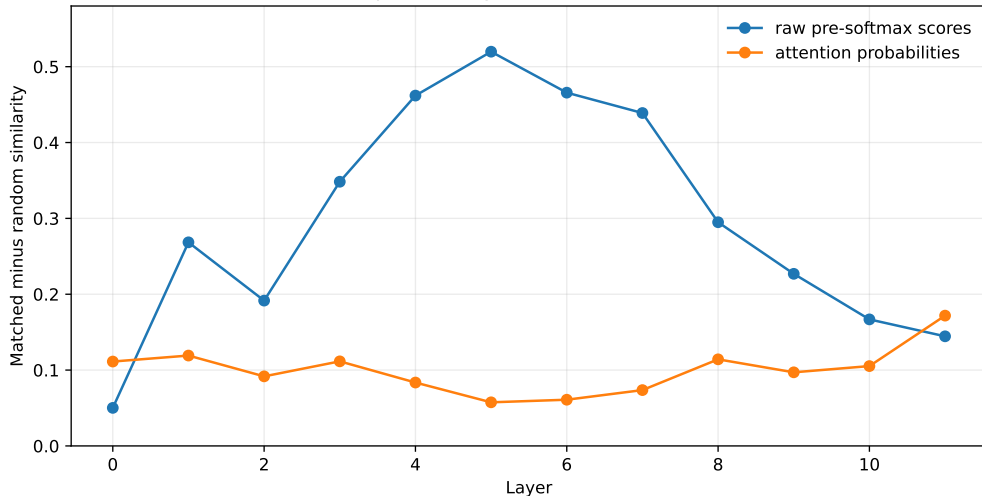
Raw pre-softmax scores are much more discriminating than returned attention probabilities.

Raw Score Layer Profile



Why Raw Scores Matter

Raw scores expose stronger relabeled cross-seed structure



- Same-index head stability is weak in many middle layers under raw-score comparison.
- Hungarian matching recovers substantially stronger cross-seed correspondence.
- This supports a weak-universality framing: similar attention-score roles emerge, but not reliably in the same head slot.
- Probability-space comparison appears too forgiving because softmax and generic attention patterns inflate similarity.

- This is still only attention-score behavior, not causal task specialization.
- The probe set is small and hand-written.
- The raw-score hook is currently implemented for GPT-NeoX/Pythia, not all model families.
- The next claim must use task-specific specialization scores, not attention similarity alone.

Use raw pre-softmax attention-score similarity as the primary Phase 0 stability metric.

Keep probability-space attention similarity as a secondary sanity check.

The next decisive experiment is to compute task-specific specialization $S(h, t)$ for simple induction / previous-token probes and compare raw vs matched cross-seed consistency.

- Script: `scripts/attention_stability.py`
- Raw-score output:
`results/phase0_pythia160m_seed1_to_9_step143000_raw_scores`
- Probability output: `results/phase0_pythia160m_seed1_to_9_step143000`
- Report: `doc/phase0_pythia160m_raw_score_pilot.md`
- Data copied into this deck folder under `data/`